



The Art of Guard

Understanding the Targeted Policy

Here's an article that's all about Allow Rules!

*S*ELinux, as mentioned in the first article of the series, is an implementation of MAC (Mandatory Access Controls).

These controls are affected through a set of rules that check the security context of the subject (e.g., the processes) and the object (e.g., the file), and allow or disallow the particular action.

There are various rules defined in an SELinux policy. To view them, use the *seinfo* command discussed earlier.

```
[root@vbg ~]# seinfo
```

Statistics for policy file: /etc/selinux/targeted/policy/policy21
Policy Version & Type: v21 (binary, MLS)

Classes:	61	Permissions:	220
Types:	1514	Attributes:	148
Users:	3	Roles:	6
Booleans:	211	Cond. Expr.:	187
Sensitivities:	1	Categories:	1024
Allow:	82576	Neverallow:	0
Auditallow:	28	Dontaudit:	5086
Role allow:	5	Role trans:	0
Type trans:	1399	Type change:	17

Type_member:	0	Range_trans:	23
Constraints:	47	Validate_trans:	0
Ps_use:	15	Genfscon:	64
Portcon:	264	Netifcon:	0
Nodecon:	8	Initial SIDs:	27

The bold text in the above output represents the information section on rules. As we can see, the default policy loaded in my system has:

- 82,756 Allow Rules
- 1,399 Type Transition Rules
- 5,086 Don't Audit Rules, and so on.

To view these rules and get an understanding of how they work, let us explore the *sesearch* command.

```
[root@vbg ~]# sesearch -a
```

```
Found 87690 av rules:
  allow bluetooth_helper_tmp_t bluetooth_helper_tmp_t :
filesystem associate ;
  allow httpd_bugzilla_script_t httpd_bugzilla_script_t :lnk_file
{ ioctl read getattr lock };
  allow avahi_t avahi_t :fifo_file { read write };
  allow avahi_t avahi_t :tcp_socket { ioctl read write create
getattr setattr append bind connect listen accept getproto se
```